

Two New Text–Tabular Datasets for MulTaBench

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Abstract

We curate and evaluate two new text–tabular datasets against the MulTaBench curation criterion, using the official implementation of five tabular learners (TabM, CatBoost, LightGBM, TabPFN v2, TabPFN-2.5) under four conditions. Both are accepted. *Vietnam housing* (30k residential listings, address text) passes on **5 of 5** learners with a joint-signal gain of $+0.288 \pm 0.024 R^2$ pooled over 25 (learner, fold) cells, positive in every cell ($t = 59.9$); *Udemy courses* (3k course listings) passes on **3 of 5** with $+0.175 \pm 0.046$, also positive in every cell ($t = 19.1$) — both two orders of magnitude above the acceptance margin $\delta = 0.001$. We report the target-awareness term with equal candour: it is the binding constraint, significant on Vietnam housing ($t = 4.7$) but not when pooled on Udemy ($t = 0.9$). We describe the mining pipeline that produced them (232 candidates screened, 2 accepted, 380 measured cells over 8.1 GPU-hours) and two screening shortcuts we measured to be invalid.

1 Introduction

Real tables are rarely purely numerical: a property listing carries an address, a course listing a title and an instructor name — free text beside numeric and categorical columns. *Text–tabular learning* asks how to use both channels at once. The question is not rhetorical: gradient-boosted trees remain very strong on structured data [10, 11], so a multimodal method only earns its complexity on tables where the text carries signal the structured block lacks, and vice versa.

That makes *dataset curation* a research problem rather than bookkeeping. A benchmark built from tables whose text merely restates their columns will report that multimodal methods do not help — a fact about the benchmark, not about the methods. MulTaBench [1] makes the requirement measurable: a table is worth benchmarking only if (i) the joint model beats *both* unimodal baselines, and (ii) a text encoder adapted to the prediction target beats a frozen one. Failing (i) collapses the task into pure tabular or pure NLP; failing (ii) exercises text encoding but not the target-aware representation learning that models such as TabSTAR [2] are built around.

We take the benchmark track: we identify, preprocess and evaluate new datasets against that criterion rather than proposing an algorithm. We contribute (1) **two new accepted datasets** — Vietnamese residential real-estate listings (5 of 5 learners) and online-course listings (3 of 5), with joint gains of $+0.288$ and $+0.175 R^2$ positive in all 25 (learner, fold) cells for both; (2) **the mining pipeline that found them** and its measured yield, 232 candidates to 2, reported stage by stage so the funnel can be sized by others; and (3) **two negative methodological results** that invalidate the obvious cheap pipeline.

2 Related Work

Tabular learning. Gradient-boosted decision trees — XGBoost [9], LightGBM [8] and CatBoost [7] — are the default for tabular prediction and remain competitive against deep architectures

Code, curation pipeline and every measured grid, log and screen: https://github.com/noabamberger/NLP_MULTABENCH

[10, 11]. Deep tabular models have closed much of the gap: FT-Transformer [6] adapted attention to heterogeneous columns, and TabM [5] reaches GBDT-level accuracy through parameter-efficient ensembling of an MLP backbone. We use LightGBM, CatBoost and TabM.

Tabular foundation models. TabPFN v2 [3] performs in-context learning over a synthetic prior, strong on small tables without per-dataset gradient training; TabPFN-2.5 [4] extends it to larger inputs. Being pretrained rather than fitted, they respond differently to a change of input representation than trees do — which is why a five-learner committee is the right instrument for a curation decision.

Text and tables together. One line of work serialises rows into sentences for a language model (TabLLM [12]); another embeds string entries as entities to transfer across tables (CARTE [13]). The line we follow keeps the tabular learner and encodes text columns into features with a contrastive sentence encoder, E5 [14], optionally adapted to the target. TabSTAR [2] makes that adaptation central — *semantically target-aware* representations — and MulTaBench operationalises it as the `ft` condition: an E5 encoder LoRA-tuned [15] on the target.

Benchmarks and curation. MulTaBench [1] is, to our knowledge, the first multimodal-tabular benchmark to publish an executable acceptance rule for its own datasets rather than a curated list alone. We reuse its reference implementation of that rule verbatim (Section 3), so our verdicts and the benchmark’s are one computation.

3 Method: the criterion, the pipeline, and the datasets

The criterion. Each dataset is evaluated with five learners — TabM, CatBoost, LightGBM, TabPFN v2 and TabPFN-2.5 — under four conditions: **Unimodal Structured** (tabular columns only), **Unimodal Unstructured** (text only), **Joint Frozen** (all features, frozen E5-small-v2 embeddings) and **Joint Target-Aware** (TAR; all features, E5 LoRA-tuned on the target). Text columns are embedded per column and compressed to 30 principal components; scores are R^2 , as both our datasets are regression tasks. Writing μ_c for the mean score over folds 0–4 in condition c , rounded to three decimals before differencing, a learner passes when

$$= \mu_{\text{frozen}} - \max(\mu_{\text{struct}}, \mu_{\text{text}}) > \delta, \quad \Delta_{\text{Aware}} = \mu_{\text{TAR}} - \mu_{\text{frozen}} > \delta, \quad (1)$$

with $\delta = 0.001$; a dataset is accepted when at least 3 of the 5 learners pass *both*, a learner clearing only one contributing nothing. Every verdict here is computed by the MulTaBench reference implementation of this rule, which asserts grid completeness (5 folds $\times$ 4 conditions per learner) rather than averaging over gaps; we did not reimplement the arithmetic.

Setup and fidelity. All grids ran on Kaggle T4 GPUs, the four conditions for a learner always measured in one session so that no environment difference can enter a delta; the four grids total 380 cells and 8.1 GPU-hours (Appendix J). We first reproduced the benchmark’s own artefacts: its published 56×10 pass matrix re-derives with **0 mismatched cells**, and on the paper’s anchor dataset the encoder-free condition reproduces published scores exactly while agrees within 0.012, every sign matching (Appendix I).

Mining pipeline. Candidates were discovered by programmatic search over public dataset repositories, then profiled, screened and gridded (Table 1); end-to-end yield is under 1%, so reaching five accepted datasets implies screening on the order of 500 candidates. For each candidate the pipeline selects a target and partitions columns without human annotation. Targets are rejected when they are identifiers or serial numbers, when one value holds more than 20% of the mass (a sentinel masquerading as a number), when $|z|_{\text{max}} > 5$, or when derived from the text they would be predicted

Table 1: Mining-funnel yield; percentages are of the 232 discovered candidates. The stage-6 screen used the flawed column filter of Appendix G, so 16 is an upper bound.

Stage	Count	Survival
Discovered by search	232	—
Parsed successfully	229	99%
Viable under typing/target gates	120	52%
After removing false text columns (dates, identifiers)	58	25%
After excluding domains already in MulTaBench	34	15%
Passing a screen	16	7%
Full grid attempted	4	—
Survived scrutiny	2	< 1%

Table 2: Per-learner results on Kaggle T4; means over folds 0–4 rounded to three decimals before differencing, $\delta = 0.001$. Struct./Text/Frozen/TAR are the four conditions of Section 3. Both grids are complete: 100 cells, no gaps.

Dataset	Learner	Struct.	Text	Frozen	TAR		Δ_{Aware}	Verdict
Vietnam housing	CatBoost	0.341	0.322	0.632	0.636	+0.291	+0.004	pass
	LightGBM	0.342	0.310	0.592	0.607	+0.250	+0.015	pass
	TabM	0.312	0.336	0.633	0.638	+0.297	+0.005	pass
	TabPFN v2	0.342	0.340	0.646	0.647	+0.304	+0.001	pass [†]
	TabPFN-2.5	0.356	0.346	0.680	0.685	+0.324	+0.005	pass
Udemy courses	CatBoost	0.297	0.210	0.480	0.501	+0.183	+0.021	pass
	LightGBM	0.261	0.212	0.479	0.474	+0.218	−0.005	fail
	TabM	0.298	0.221	0.504	0.492	+0.206	−0.012	fail
	TabPFN v2	0.299	0.162	0.435	0.437	+0.136	+0.002	pass
	TabPFN-2.5	0.311	0.211	0.452	0.464	+0.141	+0.012	pass

[†] Knife-edge: the difference of rounded means is exactly δ , clearing a strict $>$ only through float64 representation. Excluding it leaves 4 of 5, still accepted.

from; columns correlating with the target at Spearman $\rho \geq 0.95$ are dropped as leakage (Spearman, so monotone-but-nonlinear leaks such as a rank column are caught). Column typing is inferred, not declared — a non-numeric column counts as text at ≥ 100 distinct values or $\geq 80\%$ uniqueness — and it fails mainly by promoting dates and identifiers into the text block, which is what reduces 120 nominally viable candidates to 58. Two cost-saving shortcuts that look obvious are measurably invalid (Appendix G), so every positive below was re-measured with the maximal structured block and every Δ_{Aware} at ≥ 10 epochs over all five folds.

The two datasets. REG_TEXT_HOUSES_VIETNAM_2024 (nguyentiennhan/vietnam-housing-dataset-2024): 30,229 residential for-sale listings; target Price (687 distinct, $|z|_{\text{max}} = 2.54$); one text column, the free-form Address; structured block six numeric and four categorical physical attributes (area, frontage, floors, bedrooms, bathrooms, orientation, legal status, furnishing), none of which duplicates the address. REG_TEXT_EDU_UDEMY_ACADEMY (mariahalshiekh/udemy-course-academy-teaching): 2,999 online-course listings; target course price (28 distinct, $|z|_{\text{max}} = 3.00$); text columns course_name and course_instr; structured block six numeric course statistics. Both specifications were derived by the automated pipeline above — neither needed a hand-written annotation module, and neither had a column dropped for leakage.

4 Results

Vietnam housing — accepted, 5 of 5. Pooled over the 25 (learner, fold) cells (Appendix B), $= 0.288 \pm 0.024$, positive in **25 of 25**, $t = 59.9$; $\Delta_{\text{Aware}} = 0.0065 \pm 0.0068$, positive in 22 of 25, $t = 4.7$. The joint condition nearly doubles the better unimodal baseline ($0.35 \rightarrow 0.65$). The mechanism is interpretable, and is the property we think worth searching for directly: the address encodes *location premium* while the structured block encodes *physical size and condition* — orthogonal drivers of price, neither derivable from the other. The baselines are also nearly balanced (0.34 vs 0.33), the fingerprint of two independently informative channels. **One caveat we state rather than bury:** MulTaBench already contains four housing datasets. This is a different market and task, but the domain overlaps, and domain novelty is a judgement separate from whether the criterion is met.

Udemy courses — accepted, 3 of 5. $= 0.175 \pm 0.046$ pooled (Appendix C), positive in **25 of 25**, $t = 19.1$; both unimodal baselines are non-degenerate (≈ 0.30 structured, ≈ 0.21 text), so the gain is genuine complementarity rather than one modality collapsing. No course or MOOC dataset exists in MulTaBench, so the domain is genuinely new. Two weaknesses, stated plainly. Pooled over all 25 cells Δ_{Aware} is 0.0036 ± 0.0202 , $t = 0.9$ — *not* significant, and negative on two learners — though it meets the per-learner criterion on exactly the quorum of three. And we measured the dataset twice on independent machines: both lanes accept at 3 of 5 and reproduces within 0.011, but *not by the same three learners*, LightGBM and TabPFN v2 flipping in opposite directions and cancelling (Appendix D). **The dataset-level verdict is reproducible; a per-learner Δ_{Aware} on this dataset is not.**

Negative results. Of four candidates gridded, two failed (Figure A.1, Appendix E). *Board games* cleared the joint condition ($+0.047$ to $+0.059$ with the structured block restored) and died on target-awareness at 1 of 5. *Anime popularity* likewise cleared it ($+0.031$ to $+0.037$) and failed at 0 of 5; pooled, its Δ_{Aware} is -0.0012 , $t = -2.15$ — the tuned encoder is *significantly worse* than the frozen one. A Metacritic table was rejected at the target: 82% of rows carried a sentinel zero, so the regression was really a has-a-score indicator. A fifth candidate, MTG card prices, has a complete frozen grid ($= +0.050$ to $+0.075$) but no measured Δ_{Aware} and is reported as unresolved (Appendix F); screen-time rejections are listed in Appendix H. The pattern is itself a finding: joint signal is comparatively easy — 16 of 34 screened candidates cleared it — while target-awareness eliminates most.

5 Conclusion and Summary

We contribute two new text-tabular datasets that pass the MulTaBench curation criterion — one at 5 of 5 learners with a joint gain of $+0.288 R^2$ positive in every measured cell, the other at 3 of 5 with $+0.175$ — both two orders of magnitude above the acceptance margin and both measured on complete 100-cell grids with no absent cells. Equally, we report that target-awareness rather than joint signal is what eliminates candidates, that it cannot be screened cheaply along either the epoch or the fold axis, and that a name-based structured-column filter manufactures joint signal that does not exist. The most productive question when assessing a candidate is not whether it contains text, but whether the text describes a *different aspect of the entity* than the structured columns do: both accepted datasets have that property, and restatement is where the joint gain vanishes.

Limitations. *Epoch budget:* TAR fine-tuning ran 10 epochs rather than the reference default of 50 with patience 3. This is conservative — Δ_{Aware} grew monotonically with the budget in every measurement we made — so the full budget would widen our margins; it can suppress a pass, not manufacture one. *Comparability:* R^2 deltas depend on task difficulty, so the scale-free ratio of joint to best-unimodal performance ($1.85 \times$ Vietnam housing, $1.61 \times$ Udemy) transfers better across datasets.

Reproducibility: a per-learner Δ_{Aware} near the threshold need not survive a change of machine, so only dataset-level verdicts should be cited. *Scope*: two datasets, not five — the standard scope twice over but not the outstanding scope, and the measured $<1\%$ yield in Table 1 quantifies what reaching five would cost.

References

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Appendices

Result tables and one summary figure. Every number below is computed directly from the committed run CSVs under `results/curation/` in the repository above; none is transcribed by hand. Scores are R^2 . Per-state columns are means over folds 0–4 rounded to three decimals, matching the order of operations the criterion uses; pooled statistics are computed on unrounded per-(learner, fold) values.

A Outcome of every gridded (dataset, learner) cell

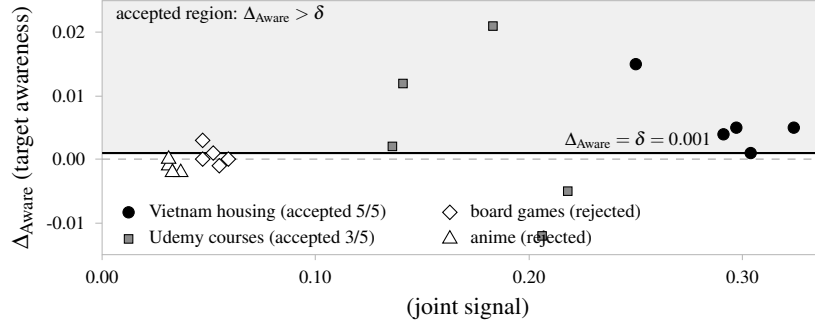


Figure A.1: Every gridded (dataset, learner) cell, against Δ_{Aware} , from the per-learner five-fold means. A learner passes only inside the shaded band. MTG cards is omitted; its Δ_{Aware} was never measured.

The horizontal spread is enormous and the vertical spread tiny. Joint signal is easy to find; target awareness is what eliminates candidates. The two rejected datasets sit far to the left on and cluster on the acceptance line on Δ_{Aware} , only one of their nine measured learners sitting clearly above it, while the two accepted ones clear it on both axes. Nothing in the figure separates accepted from rejected along the horizontal axis alone.

B Vietnam housing — complete T4 grid

Table B.1: REG_TEXT_HOUSES_VIETNAM_2024, 100 cells (5 learners $\times$ 4 states $\times$ 5 folds), one Tesla T4, ft at 10 epochs. Sources: `grid_frozen_gpu.csv`, `grid_tar.csv`, `grid_tar_pfn25.csv`.

Learner	Struct.	Text	Frozen	TAR		Δ_{Aware}	Verdict
CatBoost	0.341	0.322	0.632	0.636	+0.291	+0.004	pass
LightGBM	0.342	0.310	0.592	0.607	+0.250	+0.015	pass
TabM	0.312	0.336	0.633	0.638	+0.297	+0.005	pass
TabPFN v2	0.342	0.340	0.646	0.647	+0.304	+0.001	pass (knife-edge)
TabPFN-2.5	0.356	0.346	0.680	0.685	+0.324	+0.005	pass
5 of 5 pass; quorum 3. ACCEPTED.							

The grid is complete: zero failed cells and zero soft skips. All five learners pass both terms against a quorum of three, so the dataset is accepted. The two terms are not comparable in size. runs from +0.250 to +0.324, two orders of magnitude above δ , while Δ_{Aware} runs from +0.001 to +0.015 and is the term that could plausibly have failed. TabPFN v2 passes on a knife edge, its rounded Δ_{Aware} equal to δ ; excluding that learner leaves 4 of 5 and the verdict stands.

Table B.2: Per-(learner, fold) (left) and Δ_{Aware} (right), Vietnam housing, unrounded.

Learner	by fold					Δ_{Aware} by fold				
	0	1	2	3	4	0	1	2	3	4
CatBoost	0.2847	0.2831	0.2867	0.2924	0.2823	0.0095	0.0110	0.0056	0.0067	-0.0082
LightGBM	0.2424	0.2411	0.2501	0.2661	0.2456	0.0204	0.0188	0.0120	0.0156	0.0110
TabM	0.2989	0.2958	0.2921	0.2721	0.3137	0.0112	-0.0022	0.0056	0.0043	0.0052
TabPFN v2	0.2875	0.2998	0.3039	0.2808	0.3042	0.0063	0.0018	0.0036	0.0019	-0.0064
TabPFN-2.5	0.3103	0.3292	0.3167	0.3099	0.3097	0.0066	0.0024	0.0071	0.0102	0.0013

Pooled : $n=25$, mean 0.2880, sd 0.0240, 25/25 positive, $t = 59.88$.

Pooled Δ_{Aware} : $n=25$, mean 0.0065, sd 0.0068, 22/25 positive, $t = 4.75$, range $[-0.0082, +0.0204]$.

Pooled over the 25 cells, is 0.2880 ± 0.0240 and positive in every one ($t = 59.88$), while Δ_{Aware} is 0.0065 ± 0.0068 and positive in 22 ($t = 4.75$). The three negative Δ_{Aware} cells fall on three different learners rather than concentrating in one, so no learner’s pass rests on a systematically adverse fold. This is the pattern that distinguishes this dataset from Udemy courses: the awareness term is small here, but it is consistently signed.

Table B.3: Independent CPU frozen grid for the same dataset, 75 cells (5 learners $\times$ 3 frozen states $\times$ 5 folds). Source: `grid_frozen_cpu.csv`.

Learner	Struct.	Text	Frozen	
CatBoost	0.341	0.325	0.631	+0.290
LightGBM	0.342	0.312	0.591	+0.249
TabM	0.313	0.334	0.633	+0.299
TabPFN v2	0.342	0.340	0.645	+0.303
TabPFN-2.5	0.356	0.346	0.680	+0.324

Pooled: $n=25$, mean 0.2872, sd 0.0247, 25/25 positive, $t = 58.03$.
Agreement with the T4 lane: within 0.002 on all five learners.

This lane ran before any fine-tuning and is retained as a cross-environment control, not differenced against the GPU values. It agrees with the T4 lane to within 0.002 on all five learners, and its pooled of 0.2872 ± 0.0247 sits against 0.2880 ± 0.0240 on the T4. The joint verdict for this dataset therefore does not depend on the machine that measured it — which is not true of the awareness verdict for Udemy courses, as Appendix D shows.

C UdeMy courses — complete T4 grid

Table C.1: REG_TEXT_EDU_UDEMY_ACADEMY, 100 cells, one Tesla T4, ft at 10 epochs. Sources: `grid_gpu_light_cat_tabm.csv`, `grid_gpu_tabpfn.csv`.

Learner	Struct.	Text	Frozen	TAR		Δ_{Aware}	Verdict
CatBoost	0.297	0.210	0.480	0.501	+0.183	+0.021	pass
LightGBM	0.261	0.212	0.479	0.474	+0.218	−0.005	fail
TabM	0.298	0.221	0.504	0.492	+0.206	−0.012	fail
TabPFN v2	0.299	0.162	0.435	0.437	+0.136	+0.002	pass
TabPFN-2.5	0.311	0.211	0.452	0.464	+0.141	+0.012	pass
3 of 5 pass; quorum 3. ACCEPTED.							

Three of five learners pass, exactly the quorum, so the dataset is accepted. The split falls entirely on the awareness term: is large and positive for every learner, from +0.136 to +0.218, while Δ_{Aware} is negative on LightGBM and on TabM. CatBoost carries the widest awareness margin at +0.021 and TabPFN v2 the narrowest at +0.002. An acceptance resting on the minimum quorum with margins this thin is one we report with the qualification it deserves; the per-fold table below and Appendix D quantify it.

Table C.2: Per-(learner, fold) (left) and Δ_{Aware} (right), UdeMy courses, unrounded.

Learner	by fold					Δ_{Aware} by fold				
	0	1	2	3	4	0	1	2	3	4
CatBoost	0.1801	0.2253	0.1411	0.2115	0.1533	0.0029	0.0423	0.0131	0.0069	0.0418
LightGBM	0.2248	0.2431	0.1836	0.2238	0.2054	−0.0169	−0.0005	0.0288	−0.0479	0.0077
TabM	0.2012	0.2605	0.2252	0.1889	0.1231	−0.0221	−0.0148	−0.0189	−0.0099	0.0067
TabPFN v2	0.1318	0.1498	0.0981	0.1429	0.1583	−0.0117	0.0178	0.0068	0.0055	−0.0100
TabPFN-2.5	0.1301	0.1864	0.0863	0.1433	0.1554	0.0208	−0.0002	0.0133	0.0094	0.0193

Pooled : $n=25$, mean 0.1749, sd 0.0459, 25/25 positive, $t = 19.06$.

Pooled Δ_{Aware} : $n=25$, mean 0.0036, sd 0.0202, 15/25 positive, $t = 0.89$ (not significant), range $[-0.0479, +0.0423]$.

The contrast with Appendix B is the point of this table. is positive in all 25 cells and pools to 0.1749 ± 0.0459 ($t = 19.06$). Δ_{Aware} changes sign within a single learner across folds — LightGBM spans -0.0479 to $+0.0288$ — and pools to 0.0036 ± 0.0202 , not significant at $t = 0.89$. A per-learner awareness verdict on this dataset is therefore a statement about a margin narrower than the fold-to-fold noise, which is why only the dataset-level verdict should be cited.

D Udemy cross-environment reproduction

Table D.1: Udemy courses measured twice on independent machines, each lane a complete 100-cell grid at 10 epochs. Sources: `grid.csv` (CPU), `grid_gpu_*.csv` (T4).

Learner	CPU	T4	Δ_{Aware} CPU	Δ_{Aware} T4	Change
CatBoost	+0.194	+0.183	+0.010	+0.021	pass $\rightarrow$ pass
LightGBM	+0.209	+0.218	+0.006	−0.005	pass $\rightarrow$ fail
TabM	+0.208	+0.206	−0.007	−0.012	fail $\rightarrow$ fail
TabPFN v2	+0.136	+0.136	−0.001	+0.002	fail $\rightarrow$ pass
TabPFN-2.5	+0.140	+0.141	+0.016	+0.012	pass $\rightarrow$ pass

maximum drift 0.011; pooled CPU 0.1744 ± 0.0439 ($t = 19.85$) vs T4 0.1749 ± 0.0459 ($t = 19.06$).
 Δ_{Aware} pooled CPU 0.0044 ± 0.0178 ($t = 1.23$) vs T4 0.0036 ± 0.0202 ($t = 0.89$): neither significant.
Provenance control: `no_text` means match to three decimals across lanes for all five learners (0.297/0.261/0.298/0.311/0.299). That state touches no encoder, so loading, curation, the 90/10 split, the seeding and the learners reproduce exactly; divergence enters only with text.

Both lanes accept at 3 of 5, but not with the same three learners. LightGBM passes on CPU and fails on the T4, TabPFN v2 does the reverse, and the two flips cancel to leave the dataset-level verdict intact. The joint term is stable across the change of machine, drifting by at most 0.011, and neither pooled Δ_{Aware} is significant in either lane. The provenance rows at the foot of the table locate where the divergence enters: the encoder-free `no_text` state matches to three decimals across lanes on all five learners, so loading, curation, the 90/10 split, the seeding and the learners themselves are not the source. Only the text path differs.

E Rejected candidates — complete grids

Table E.1: Board games (`melissamonfared/board-games`), 100 cells on one T4, measured with the *maximal* structured block restored.

Learner	Struct.	Text	Frozen	TAR		Δ_{Aware}	Verdict
CatBoost	0.576	0.490	0.623	0.623	+0.047	0.000	fail
LightGBM	0.569	0.477	0.616	0.619	+0.047	+0.003	pass
TabM	0.576	0.504	0.628	0.629	+0.052	+0.001	knife-edge (counted fail)
TabPFN v2	0.585	0.504	0.640	0.639	+0.055	−0.001	fail
TabPFN-2.5	0.586	0.514	0.645	0.645	+0.059	0.000	fail

1 of 5 counted honestly (2 of 5 nominally); quorum 3. REJECTED.
Pooled : mean 0.0523, sd 0.0072, 25/25 positive, $t = 36.21$.
Pooled Δ_{Aware} : mean 0.0003, sd 0.0058, 10/25 positive, $t = 0.25$, range $[-0.0124, +0.0163]$.

is healthy and Δ_{Aware} is not. Every learner clears the joint term, which pools to 0.0523 ± 0.0072 and is positive in all 25 cells, but the awareness term pools to 0.0003 ± 0.0058 with only 10 of 25 cells positive and $t = 0.25$ — indistinguishable from zero. TabM’s Δ_{Aware} of +0.001 equals δ after rounding and is counted as a fail, which is why the table reads 1 of 5 counted honestly against 2 of 5 nominally. Either count falls short of the quorum. This candidate is the clearest instance of the pattern in Figure A.1: a real joint gain that the awareness term nonetheless rejects.

Table E.2: Anime popularity (wiltheman/anime-data-set-for-ml, 10,000 rows, target popularity, text = genres, themes, title_english; members and scored_by dropped as leakage). 80 cells on one T4.

Learner	Struct.	Text	Frozen	TAR		Δ_{Aware}	Verdict
CatBoost	0.821	0.441	0.852	0.851	+0.031	-0.001	fail
LightGBM	0.816	0.435	0.847	0.847	+0.031	0.000	fail
TabM	0.816	0.442	0.853	0.851	+0.037	-0.002	fail
TabPFN v2	0.826	0.455	0.859	0.857	+0.033	-0.002	fail
TabPFN-2.5	not run						non-pass

0 of 5; quorum 3. REJECTED.
Pooled : $n=20$, mean 0.0329, sd 0.0066, 20/20 positive, $t = 22.31$.
Pooled Δ_{Aware} : $n=20$, mean -0.0012, sd 0.0026, 7/20 positive, $t = -2.15$.

TabPFN-2.5 was not run, so its cell counts as a non-pass and the quorum denominator stays at five; the dataset fails at 0 of 5. As with board games the joint term is comfortable, pooling to 0.0329 ± 0.0066 and positive in all 20 measured cells. What sets this candidate apart is the direction of the awareness term: Δ_{Aware} pools to -0.0012 at $t = -2.15$, so the tuned encoder is significantly *worse* than the frozen one. This is the only candidate we gridded where adapting the encoder to the target actively hurt.

Table E.3: Metacritic PC games — rejected at the target.

Quantity	Value
Source rows	12,624
Rows with sentinel <code>Metacritic = 0</code> (“never reviewed”)	10,357 (82%)
Scored rows after repair	2,267 (20–96, 71 distinct, $ z _{\text{max}} = 4.67$)
Surviving text columns	1 (<code>ResponseName</code> , a title — a proper noun, not free text)
Structured columns	15, dominated by <code>RecommendationCount</code> (importance 0.62)
Grid completeness	52 rows, 3 frozen states, no <code>ft</code> state
Fold-0 Δ_{Aware} screen	-0.014 (LightGBM), -0.006 (CatBoost) — wrong sign on both

The rejection happened before a verdict was worth computing. A sentinel `Metacritic` score of zero, meaning never reviewed, covers 82% of rows, so the nominal regression is really a has-a-score indicator; repairing it leaves 2,267 scored rows. The only surviving text column is a title, a proper noun rather than free text, and the fold-0 Δ_{Aware} screen comes out the wrong sign on both learners tried. The partial grid is reported for completeness and no verdict is claimed from it.

F MTG card prices — frozen grid only, unresolved

Table F.1: REG_TEXT_GAMES_MTG_CARD_PRICES, 75/75 frozen cells on CPU. Target is $\log_{10}(\text{USD})$.

Learner	Struct.	Text	Frozen	
CatBoost	0.525	0.291	0.587	+0.062
LightGBM	0.518	0.265	0.575	+0.057
TabM	0.505	0.308	0.580	+0.075
TabPFN v2	0.521	0.270	0.571	+0.050
TabPFN-2.5	0.523	0.310	0.591	+0.068
Pooled : $n=25$, mean 0.0624, sd 0.0112, 25/25 positive, $t = 27.81$.				
Δ_{Aware} : not measured . Requires ≥ 10 epochs over ≥ 3 folds (Appendix G).				

The joint term is not in doubt: is positive on all five learners and in all 25 cells, pooling to 0.0624 ± 0.0112 at $t = 27.81$. Δ_{Aware} has never been run, so the dataset is reported as unresolved rather than as an acceptance; resolving it requires at least 10 epochs over at least three folds, for the reasons established in Appendix G. The target also carries heavy tails. The raw price column has $|z|_{\max} = 103$, and even after the log transform $|z|_{\max} = 5.36$ — warned but never clipped, per the reference protocol.

G Validity evidence: the two invalid shortcuts

Two obvious ways to cut curation cost are measurably invalid, each because it collapses the quantity being measured. **(1) Never delete structured columns by name.** A junk filter dropping columns that match year or time removes Year Published and Play Time from a board-games candidate — not identifiers but its two strongest structured predictors — letting the text proxy for them and inventing a joint gain that vanishes on restoration. The unstructured baseline never moves, so this is a controlled ablation rather than a correlation; name patterns are therefore used only to reject *text* candidates. **(2) Δ_{Aware} cannot be screened cheaply.** A cheap screen is valid only where cheapness does not change the quantity screened. That holds for (frozen encoders, so a fold subset is unbiased) and fails for Δ_{Aware} on two independent axes: an under-trained adapter drives TAR toward frozen by construction, and a single fold cannot resolve a threshold sitting deep inside the per-(learner, fold) noise band. Screening on one fold cost two full grids.

Table G.1: **Axis 1 — structured-column deletion manufactures** . Board games, LightGBM, fold 0, target Complexity Average, text = Mechanics + Name.

Specification	Struct.	Text	Frozen	
auto-spec (Year Published, Play Time deleted)	0.584	0.502	0.623	+0.0388
Year Published restored	0.613	0.502	0.636	+0.0229
full structured block restored	0.684	0.502	0.684	−0.0005
same, semantically natural target (Rating Average)	0.567	0.335	0.567	+0.0006

Columns are restored one at a time. The unstructured baseline never moves from 0.502, so the entire effect sits on the structured baseline — which is what makes this a controlled ablation rather than a correlation. Deleting Year Published and Play Time yields $= +0.0388$; restoring Year Published alone cuts that to $+0.0229$; restoring the full block drives it to $−0.0005$. The apparent joint gain was an artefact of the deletion at every step, and a semantically natural target behaves the

same way at +0.0006. Name patterns are therefore used in this pipeline only to reject *text* candidates, never to drop structured columns.

Table G.2: **Axis 2a — the epoch budget.** Identical cells, Udemý, fold 0, at two and at ten epochs.

Learner (fold 0)	Frozen	TAR @ 2 ep	Δ_{Aware} @ 2 ep	TAR @ 10 ep	Δ_{Aware} @ 10 ep
LightGBM	0.4957	0.5056	+0.0099	0.5279	+ 0.0322
CatBoost	0.5177	0.5282	+0.0105	0.5299	+ 0.0122
At 2 epochs the full Udemý sweep returned 1 of 5 — an artefact of the budget, not of the dataset.					

An under-trained LoRA adapter leaves TAR close to frozen, collapsing Δ_{Aware} toward zero by construction. Raising the budget from 2 to 10 epochs more than triples LightGBM’s Δ_{Aware} , from +0.0099 to +0.0322, and moves CatBoost’s from +0.0105 to +0.0122. At 2 epochs the full Udemý sweep returned 1 of 5 — an artefact of the budget, not a property of the dataset. A *fail* at a low budget therefore proves nothing; only a pass is informative. The same asymmetry is why the 10-epoch budget used throughout can suppress an acceptance but cannot manufacture one.

Table G.3: **Axis 2b — the fold count.** Fold-0 Δ_{Aware} against the five-fold mean on the two gridded rejections.

Learner	Board games			Anime		
	fold 0	5-fold	sign	fold 0	5-fold	sign
CatBoost	+0.0074	−0.0003	flip	+0.0018	−0.0008	flip
LightGBM	+0.0163	+0.0021	—	+0.0034	−0.0005	flip
TabM	+0.0052	+0.0010	—	+0.0010	−0.0016	flip
TabPFN v2	+0.0010	−0.0015	flip	−0.0041	−0.0020	—
TabPFN-2.5	+0.0064	+0.0001	—	not run		
Per-(learner, fold) spread: board games $\sigma = 0.0058$ over $[-0.0124, +0.0163]$; anime $\sigma = 0.0026$ over $[-0.0066, +0.0034]$. The threshold $\delta = 0.001$ sits inside both bands, so one fold cannot resolve the criterion. Screening on one fold cost two full grids.						

Every board-games learner drops between the two columns and two change sign; anime’s positive fold-0 reading reverses entirely, with three of its four measured learners flipping. The per-(learner, fold) spread quoted at the foot of the table puts $\delta = 0.001$ well inside both noise bands, so a single fold cannot resolve the criterion in either direction. We screened on one fold before establishing this, and it cost two full grids.

H Screen-time rejections

Table H.1: Candidates rejected before a full grid was spent, each with the measured quantity that condemns it.

Candidate	Reason (with the measured quantity that condemns it)
steam-games-dataset-steamspyauto-target	was appid, a primary key; no free text besides proper nouns
gog-com-video-games	auto-target was a date; re-screened $= -0.0003$, text $R^2 = 0.996$ (the text <i>was</i> the release dates)
popular-video-games-1980-2025	512 rows; no numeric column qualified as a target
lego-sets	12,261 rows are 744 products replicated across 21 countries — near-duplicates straddle train and test
top-women-chess-players	$= +0.024$ inside the ± 0.015 band, text $R^2 = 0.074$ (text nearly inert)
kerala-bevco-liquor-price	auto-target was S1 No; with the corrected target, structured $R^2 = 1.000$ (deterministic)
google-play-store-apps	target derived from its own text
daraz-perfumes	every state negative ($-0.130 / -0.009 / -0.045$): a delta between them is meaningless
global-grocery-nutrition-2025	$= +0.0005$ with structured $R^2 = 0.969$ — saturated, no headroom
global-movies-1950-2026	target roi_pct = revenue/budget, an arithmetic identity over two structured columns; structured $R^2 = 0.997$
steam-games-dataset	no numeric column qualified as a target
ultimate-games-15k	auto-target was serial_no; $= -0.0027$ against structured $R^2 = 0.983$
horror-movies-profits	healthy ($+0.055 / +0.093$) but fold-0 $\Delta_{\text{Aware}} -0.001 / -0.013$
vgsales-userscore	every delta at noise scale: $+0.001/+0.004$, $\Delta_{\text{Aware}} +0.001/+0.002$

Two classes recur. The first is a junk target chosen automatically: appid, S1 No and serial_no are primary keys or serial numbers, and one candidate had a release date selected as its target. The second is a saturated or degenerate baseline against which no text signal can be demonstrated, with structured R^2 reaching 0.969, 0.983, 0.997 and 1.000 on four separate candidates. Both classes are cheap to detect from a profile of the target and the structured block, and neither needs a grid to rule out — which is what makes the profiling stage of the funnel worth its cost.

I Harness validation against the benchmark’s own anchor

Table I.1: Anchor dataset MUL_TEXT_PRODUCT_SENTIMENT ($N = 5,091$, 4 classes, 1 text + 1 structured column); 36-run frozen grid.

Check	Result
Published 56×10 pass matrix re-derived from released scores	0 mismatched cells
Verdict function on a known accepted dataset	accepted, 5 of 5
Verdict function on a known rejected dataset	rejected, 0 of 5
no_text (encoder-free) vs published, mean absolute difference	0.0000
LightGBM no_text, fold 0	0.83454303717305 (identical)
text_only vs published, mean signed difference	-0.0000 (range -0.046 to $+0.038$)
all vs published, mean signed difference	$+0.0058$ (max $+0.0235$)
Frozen-embedding cache, cached vs uncached score	0.8618478948517495, bit-identical

These checks ran before any candidate number was trusted. The published 56×10 pass matrix re-derives from the released scores with zero mismatched cells, and the verdict function returns the published outcome on both a known accepted and a known rejected dataset. The encoder-free no_text condition reproduces published scores exactly — to fourteen decimal places on the LightGBM fold-0 cell — which pins loading, curation, splitting and seeding. The encoder-dependent conditions agree less tightly, all carrying a $+0.0058$ mean signed difference. That gap is the expected consequence of a different embedding environment, not of a different protocol, and it is why the next table validates itself rather than the raw scores it is built from.

Table I.2: — the quantity the criterion consumes — reproduced on the anchor.

Learner	(ours)	(published)	Sign agrees
CatBoost	0.075	0.073	yes
LightGBM	0.058	0.046	yes
TabM	0.085	0.086	yes
TabPFN v2	0.076	0.079	yes

The deltas agree within 0.012 across four learners with every sign agreeing, and three of the four agree within 0.003. The single wider case is LightGBM, at 0.058 against a published 0.046. This reproduction is the calibration behind the ± 0.015 fold-noise band used throughout the paper: a candidate whose falls inside that band is not distinguishable from measurement noise, and is screened out rather than gridded.

J Compute cost

Table J.1: Measured GPU time, summed over the per-cell timings recorded in the run CSVs.

Dataset	Rows	Cells	GPU-hours	Outcome
Udemy courses	2,999	100	0.59	accepted, 3 of 5
Vietnam housing	30,229	100	1.87	accepted, 5 of 5
Board games	20,343	100	3.56	rejected, 1 of 5
Anime popularity	10,000	80	2.05	rejected, 0 of 5
Total		380	8.07	2 accepted

All four conditions for a given learner share one session, so no environment difference can enter a delta. The four gridded candidates cost 8.07 GPU-hours over 380 cells in total. Cost is not proportional to row count: evaluation subsamples to at most 10k training rows, so the remaining variation is driven by text length and by the learner mix rather than by dataset size. Vietnam housing has $1.5\times$ the rows of board games and costs half as much. The practical consequence for anyone sizing this work is that the grid is the cheap part — under two GPU-hours for an accepted dataset — and the screening that precedes it is where the effort goes.

Table J.2: Optimisations that made the grids affordable.

Optimisation	Speed-up	Result-preserving?
Per-string frozen embedding cache	$\approx 40\times$	yes, bit-exact (asserted by tests)
TAR encoder sharing (25 runs $\rightarrow$ 10 fine-tunings)	$2.5\times$	yes
Dynamic max_length cap, frozen encode	$\approx 7\times$	no ($\approx 10^{-7}$ drift)
Dynamic max_length cap, TAR training step	333s $\rightarrow$ 8s	no ($\approx 10^{-7}$ drift)

Only the result-preserving optimisations are used for numbers compared against published values. The frozen-embedding cache is the largest single win, roughly $40\times$ and bit-exact, asserted by tests. The two `max_length` caps are not exact, but their drift is on the order of 10^{-7} — four orders of magnitude below δ — so they cannot move a verdict.